

Project structure

```
stt/
├── script.py
├── vendor/
├── models/
│   └── vosk-model-small-en-us-0.15/
│       ├── am/
│       ├── conf/
│       ├── graph/
│       └── ivec/
```

Install system packages:

```
mkdir -p vendor
python3 -m pip install --target ./vendor sounddevice vosk openwakeword numpy
sudo ldconfig
```

Install Python packages locally:

```
sudo apt update
sudo apt install python3-pip libportaudio2 portaudio19-dev unzip wget
sudo ldconfig
```

Download the Vosk model:

```
mkdir -p models
cd models
wget https://alphacephei.com/vosk/models/vosk-model-small-en-us-0.15.zip
unzip vosk-model-small-en-us-0.15.zip
cd ..
```

List input devices, then use whichever you get or want:

```
PYTHONPATH=./vendor python3 - <<'PY'
import sounddevice as sd
for i, dev in enumerate(sd.query_devices()):
    if dev["max_input_channels"] > 0:
        print(i, dev["name"], "inputs=", dev["max_input_channels"])
PY
```

Run the script:

(Example using microphone device 1)

```
python3 script.py \
--vosk-model ./models/vosk-model-small-en-us-0.15 \
--device 1 \
--sample-rate 48000 \
--wakeword-name alexa \
```